

# Task 1 — Student Management Database

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## 1. Database Schema Design

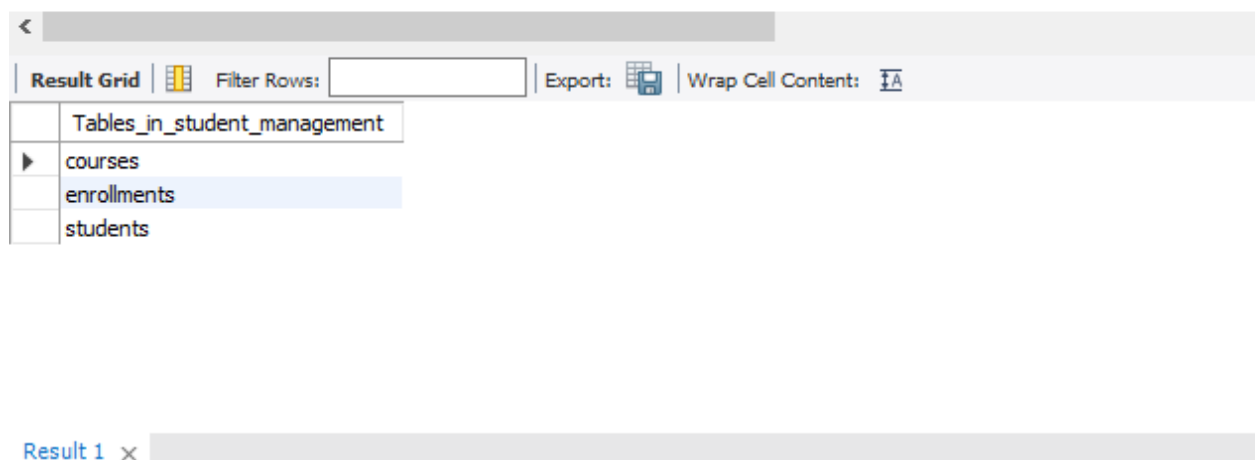
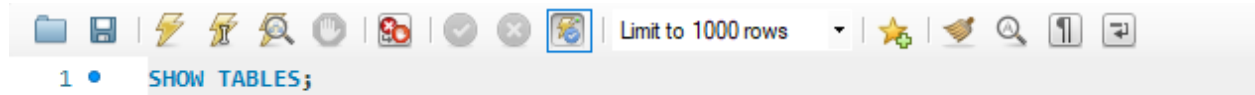
A Student Management System manages:

- Students
- Courses
- Student Enrollments

The database contains 3 tables:

1. Students
2. Courses
3. Enrollments

## 2. Tables Creation



## 3. Appropriate Columns and Data Types

### Students Table

| Column Name | Data Type    | Description                |
|-------------|--------------|----------------------------|
| student_id  | INT          | Unique ID for each student |
| name        | VARCHAR(100) | Student name               |
| age         | INT          | Student age                |
| gender      | VARCHAR(10)  | Gender                     |
| email       | VARCHAR(100) | Student email              |
| phone       | VARCHAR(15)  | Phone number               |

### Courses Table

| Column Name | Data Type | Description               |
|-------------|-----------|---------------------------|
| course_id   | INT       | Unique ID for each course |

|             |               |                 |
|-------------|---------------|-----------------|
| course_name | VARCHAR(100)  | Name of course  |
| duration    | VARCHAR(50)   | Course duration |
| fees        | DECIMAL(10,2) | Course fees     |

## Enrollments Table

| Column Name     | Data Type | Description          |
|-----------------|-----------|----------------------|
| enrollment_id   | INT       | Unique enrollment ID |
| student_id      | INT       | Student reference    |
| course_id       | INT       | Course reference     |
| enrollment_date | DATE      | Date of enrollment   |

## 4. Primary Keys and Foreign Keys

### Primary Keys

| Table       | Primary Key   |
|-------------|---------------|
| Students    | student_id    |
| Courses     | course_id     |
| Enrollments | enrollment_id |

### Foreign Keys

| Table       | Foreign Key | References           |
|-------------|-------------|----------------------|
| Enrollments | student_id  | Students(student_id) |
| Enrollments | course_id   | Courses(course_id)   |

## Task 2 — SQL Table Creation

### 1. Create Students Table



```
CREATE TABLE Students (  
    student_id INT PRIMARY KEY,  
    name VARCHAR(100),  
    age INT,  
    gender VARCHAR(10),  
    email VARCHAR(100),  
    phone VARCHAR(15)  
);
```

## 2. Create Courses Table



```
CREATE TABLE Courses (  
    course_id INT PRIMARY KEY,  
    course_name VARCHAR(100),  
    duration VARCHAR(50),  
    fees DECIMAL(10,2)  
);
```

## 3. Create Enrollments Table

```

CREATE TABLE Enrollments (
    enrollment_id INT PRIMARY KEY,
    student_id INT,
    course_id INT,
    enrollment_date DATE,

    FOREIGN KEY (student_id)
    REFERENCES Students(student_id),

    FOREIGN KEY (course_id)
    REFERENCES Courses(course_id)
);

```

## Primary Keys Used

| Table       | Primary Key   |
|-------------|---------------|
| Students    | student_id    |
| Courses     | course_id     |
| Enrollments | enrollment_id |

## Foreign Keys Used

| Table       | Foreign Key | References           |
|-------------|-------------|----------------------|
| Enrollments | student_id  | Students(student_id) |
| Enrollments | course_id   | Courses(course_id)   |

## OUTPUT

- 1 • `SELECT * FROM Students;`
- 2 • `SELECT * FROM Courses;`
- 3 • `SELECT * FROM Enrollments;`

<   Filter Rows:  | Edit:   | Export/Import

|   | student_id | name    | age  | gender | email             | phone      |
|---|------------|---------|------|--------|-------------------|------------|
| ▶ | 1          | Vagdevi | 21   | Female | vagdevi@gmail.com | 9876543210 |
|   | 2          | Rahul   | 22   | Male   | rahul@gmail.com   | 9876501234 |
|   | 3          | Sneha   | 20   | Female | sneha@gmail.com   | 9876512345 |
| * | NULL       | NULL    | NULL | NULL   | NULL              | NULL       |

Result Grid   Filter Rows:  | Edit:  

|   | course_id | course_name     | duration | fees     |
|---|-----------|-----------------|----------|----------|
| ▶ | 101       | Python          | 3 Months | 15000.00 |
|   | 102       | Java            | 4 Months | 20000.00 |
|   | 103       | Web Development | 6 Months | 25000.00 |
| * | NULL      | NULL            | NULL     | NULL     |

Result Grid   Filter Rows:  | Edit:  

|   | enrollment_id | student_id | course_id | enrollment_date |
|---|---------------|------------|-----------|-----------------|
| ▶ | 1             | 1          | 101       | 2026-05-13      |
|   | 2             | 2          | 102       | 2026-05-13      |
|   | 3             | 3          | 103       | 2026-05-13      |
| * | NULL          | NULL       | NULL      | NULL            |

## Task 3 — Database Relationships

| Entity Relationship | Relationship Type  | Explanation                                                                  |
|---------------------|--------------------|------------------------------------------------------------------------------|
| Teacher → Students  | One-to-Many (1:M)  | One teacher can teach many students                                          |
| User → Passport     | One-to-One (1:1)   | One user has one passport                                                    |
| Students → Courses  | Many-to-Many (M:N) | One student can enroll in many courses and one course can have many students |
| Customer → Orders   | One-to-Many (1:M)  | One customer can place many orders                                           |

## Task 4 — Real-World Database Schema

### Selected Application:

### Online Shopping System

An Online Shopping System manages:

- Customers
- Products
- Orders
- Payments

# Database Schema Design

Table 1 — Customers

| Column Name | Data Type    | Description        |
|-------------|--------------|--------------------|
| customer_id | INT          | Unique customer ID |
| name        | VARCHAR(100) | Customer name      |
| email       | VARCHAR(100) | Customer email     |
| phone       | VARCHAR(15)  | Phone number       |
| address     | VARCHAR(200) | Customer address   |

Primary Key

customer\_id

Table 2 — Products

| Column Name  | Data Type     | Description        |
|--------------|---------------|--------------------|
| product_id   | INT           | Unique product ID  |
| product_name | VARCHAR(100)  | Product name       |
| price        | DECIMAL(10,2) | Product price      |
| stock        | INT           | Available quantity |

Primary Key

product\_id



## Table 3 — Orders

| Column Name  | Data Type     | Description        |
|--------------|---------------|--------------------|
| order_id     | INT           | Unique order ID    |
| customer_id  | INT           | Customer reference |
| order_date   | DATE          | Order date         |
| total_amount | DECIMAL(10,2) | Total order amount |

### Primary Key

order\_id

### Foreign Key

customer\_id → Customers(customer\_id)

## Table 4 — Payments

| Column Name    | Data Type   | Description       |
|----------------|-------------|-------------------|
| payment_id     | INT         | Unique payment ID |
| order_id       | INT         | Order reference   |
| payment_method | VARCHAR(50) | Payment type      |
| payment_status | VARCHAR(50) | Payment status    |

### Primary Key

payment\_id

### Foreign Key

order\_id → Orders(order\_id)

## Relationships Between Tables

| Relationship       | Type         |
|--------------------|--------------|
| Customers → Orders | One-to-Many  |
| Orders → Payments  | One-to-One   |
| Products → Orders  | Many-to-Many |

## SQL CREATE TABLE Queries

### 1. Customer table

```
CREATE TABLE Customers (
    customer_id INT PRIMARY KEY,
    name VARCHAR(100),
    email VARCHAR(100),
    phone VARCHAR(15),
    address VARCHAR(200)
);
```

### 2. Products table

```
CREATE TABLE Products (
    product_id INT PRIMARY KEY,
    product_name VARCHAR(100),
    price DECIMAL(10,2),
    stock INT
);
```

### 3. Orders Table

```
CREATE TABLE Orders (  
    order_id INT PRIMARY KEY,  
    customer_id INT,  
    order_date DATE,  
    total_amount DECIMAL(10,2),  
  
    FOREIGN KEY (customer_id)  
    REFERENCES Customers(customer_id)  
);
```

#### 4. Payments Table

```
CREATE TABLE Payments (  
    payment_id INT PRIMARY KEY,  
    order_id INT,  
    payment_method VARCHAR(50),  
    payment_status VARCHAR(50),  
  
    FOREIGN KEY (order_id)  
    REFERENCES Orders(order_id)  
);
```

## TABLES CREATED:

- 1 • `SELECT * FROM Customers;`
- 2 • `SELECT * FROM Products;`
- 3 • `SELECT * FROM Orders;`
- 4 • `SELECT * FROM Payments;`

<   Filter Rows:  | Edit:    | Export

|   | customer_id | name    | email             | phone      | address   |
|---|-------------|---------|-------------------|------------|-----------|
| ▶ | 1           | Vagdevi | vagdevi@gmail.com | 9876543210 | Hyderabad |
|   | 2           | Rahul   | rahul@gmail.com   | 9123456780 | Bangalore |
|   | 3           | Sneha   | sneha@gmail.com   | 9988776655 | Chennai   |
| * | NULL        | NULL    | NULL              | NULL       | NULL      |

**Result Grid**   Filter Rows:  | Edit: 

|   | product_id | product_name | price    | stock |
|---|------------|--------------|----------|-------|
| ▶ | 101        | Laptop       | 65000.00 | 10    |
|   | 102        | Mobile Phone | 25000.00 | 20    |
|   | 103        | Headphones   | 2000.00  | 50    |
| * | NULL       | NULL         | NULL     | NULL  |

**Result Grid**   Filter Rows:  | Edit: 

|   | order_id | customer_id | order_date | total_amount |
|---|----------|-------------|------------|--------------|
| ▶ | 1001     | 1           | 2026-05-13 | 67000.00     |
|   | 1002     | 2           | 2026-05-13 | 25000.00     |
|   | 1003     | 3           | 2026-05-13 | 2000.00      |
| * | NULL     | NULL        | NULL       | NULL         |

|             |                                                                                   |                                                                                   |                                   |                                                                                         |                                                                                   |                                                                                   |
|-------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------|-----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Result Grid |  |  | Filter Rows: <input type="text"/> | Edit:  |  |  |
|             | payment_id                                                                        | order_id                                                                          | payment_method                    | payment_status                                                                          |                                                                                   |                                                                                   |
| ▶           | 5001                                                                              | 1001                                                                              | Credit Card                       | Completed                                                                               |                                                                                   |                                                                                   |
|             | 5002                                                                              | 1002                                                                              | UPI                               | Completed                                                                               |                                                                                   |                                                                                   |
|             | 5003                                                                              | 1003                                                                              | Cash on Delivery                  | Pending                                                                                 |                                                                                   |                                                                                   |
| ✱           | NULL                                                                              | NULL                                                                              | NULL                              | NULL                                                                                    |                                                                                   |                                                                                   |